

8.5 A Command-Aware Hybrid LDO for Advanced HBM Interfaces with 150μA Quiescent Current and 20pF On-Chip Capacitor Achieving Sub-10mV Voltage Droop in 400ps Settling Time

Publisher: IEEE Cite This

<< Results | < Previous | Next >

Jaeho Kim ; Myeongho Han ; Jooeun Bang ; Younghyun Lim ; Jaehyouk Choi ...

6

Cites in Papers

3253

Full Text Views

Abstract
Authors
Figures
References
Citations
Keywords
Metrics
More Like This



Downr
PDF

Abstract:
With the advent of the generative AI era, high-bandwidth memory (HBM) has emerged as an irreplaceable solution that can provide ultra-high memory bandwidth (BW) of more t... [Show More](#)

▼ Metadata

≡ Contents

An LDO is a common solution to the PSIJ problem, but its requirements for the WDQS buffer are challenging. First, the amount of voltage droop (ΔV_{DROOP}) caused by the WDQS toggle must be reduced to sufficiently small (e.g., <10mV) in an extremely short settling time (T_s). T_s should be reduced to less than 400ps when $\text{WDQS}/2_{\text{IN}}$ toggles at 2.5GHz since the write-preamble time (t_{WPRED1}) before the first DQ data is only 4UI (or 1 cycle of $\text{WDQS}/2_{\text{IN}}$) [1]. Second, power-supply rejection (PSR) must be provided across all frequencies to protect the WDQS buffer from external noise. Third, these goals must be achieved with minimal power, area, and capacitors. The right of Fig. 8.5.1 shows that, to reduce both ΔV_{DROOP} and T_s with a small output capacitor (C_{OUT}), analog LDOs (ALDOs) [2]– [6] require a wide-BW amplifier and thus a huge quiescent current (I_0), but achieving such an extremely short T_s within the given power budget is very difficult. As an alternative to conventional digital LDOs [7]– [9], which require higher clock frequencies for smaller T_s , event-driven LDOs (ED-LDOs) with digital [10]– [18] or hybrid [19]– [21] architectures can eliminate this tradeoff by using an adaptive clock that increases frequency only when an event occurs. However, responding immediately to a random event is challenging and unreliable, leading to an extended response time (T_R) and inevitably increasing ΔV_{DROOP} .

This work presents a command-aware hybrid LDO (CA-HLDO) that can minimize ΔV_{DROOP} and T_s while using minimal I_Q , area, and capacitors. To achieve a breakthrough that can overcome the limitations of existing approaches, we propose a method that exploits the unique property of memory: it always performs predefined operations according to commands from the host. The proposed CA-HLDO leverages this property to supply the WDQS buffer with the optimal I_L at the precise timing derived from ACT_{CMD} and $\text{WDQS}/2_{\text{IN}}$, thereby minimizing the PSIJ of $S_{0\text{UTXS}}$. The bottom left of Fig. 8.5.1 shows that the CA-HLDO consists of (1) a DLDO that instantaneously injects an optimally adapted digital current (I_D) and (2) a low-power ALDO that compensates for the residual I_L with an analog current (I_A). Conventional ED-LDOs wait for unpredictable random events, so they inherently experience long T_R . However, the proposed CA-HLDO prepares the optimal I_D in advance and provides it immediately when ACT_{CMD} and $\text{WDQS}/2_{\text{IN}}$ toggle,



minimizing both T_R and T_S (bottom right of Fig. 8.5.1). The CA-HLDO also can provide at least -10dB PSR across all frequencies while consuming only $150\mu\text{A}/Q$ and $20\text{pF } C_{\text{OUT}}$.

Figure 8.5.2 shows the overall architecture of the proposed CA-HLDO. The DLDO has two separate PMOS pass-transistor switch arrays, i.e., $M_{P,ACT}$ and $M_{P,TOG}$, that supply two independent I_D s, i.e., $I_{D,ACT}$ and $I_{D,TOG}$, respectively, which are adjusted continuously in the background by the I_D -adaptor to accurately compensate for the I_L of the WDQS buffer. The control-signal generator (CSG) provides signals to operate the I_D -adaptor and $M_{P,ACT}/M_{P,TOG}$. The bottom left of Fig. 8.5.2 shows that the WDQS buffer consists of a series of sub-components, i.e., a CML buffer, a CML-to-CMOS (C2C) buffer, and an inverter chain. The profile of I_L from this WDQS buffer is shown at the top of Fig. 8.5.3, and it can be divided into four sections ($S1 - S4$) based on the sequential operations of the sub-components. In the first section ($S1$) when ACT_{CMD} is low, the WDQS buffer is in standby mode with nearly zero I_L . After ACT_{CMD} goes high, in $S2$, the CML buffer is activated and begins drawing a static current ($I_{L,ACT}$) of approximately 2mA from the input voltage (V_{IN}), causing the first ΔV_{DR00F} at the LDO output voltage, V_{OUT} . The CSG generates s_{AC1} synchronously at the rising edge of ACT_{CMO} which instantaneously turns on $M_{P,AC1}$, generating $I_{D,ACT}$ to compensate for $I_{L,ACT}$. A few nanoseconds later, the quadrature WDQS/ 2_{IN} starts toggling at 2.5GHz in $S3$. This sudden toggle momentarily generates $I_{L,TOG}$ of approximately 12mA through the C2C buffer (0.2mA) and the following inverter chain (11.8mA), causing an even larger ΔV_{DROOP} . The CSG instantly generates S_{TOG} , which goes high during every toggle of WDQS/ 2_{IN} , to turn on $M_{P,TOG}$, generating $I_{D,TOG}$ to compensate for $I_{L,TOG}$. Since S_{TOG} is generated by the two output signals of the C2C buffer (C2C, and C2C_Q), the timings of $I_{L,TOG}$ and $I_{D,TOG}$ are synchronized, thereby making T_R nearly zero. As soon as WDQS/ 2_{IN} stops toggling in $S4$, $I_{L,TOG}$ and $I_{D,TOG}$ cease simultaneously, preventing an overshoot at V_{OUT} . Since most of I_L is supplied by the DLDO, the role of I_A of the ALDO is limited to compensating for the residual I_L , allowing the ALDO to have a small BW with minimal I_Q . The error amplifier was designed based on a cross-coupled amplifier [6], [22] with a local feedforward path to improve stability [23].

Due to a small C_{OUT} , the ALDO has the dominant pole at the gate of its pass transistor (M_{PA}), resulting in an inevitable high-frequency PSR hump (bottom right of Fig. 8.5.2). Generally, the DLDO has poor PSR, so a native NMOS transistor with an RC filter (R_1 and C_1) was used at the bottom of each $M_{P,AC1}$ and $M_{P,TOG}$ to filter the high-frequency supply noise [24–26]. The PSRs of the ALDO and the DLDO complement each other, enhancing the low-frequency PSR of the DLDO and the high-frequency PSR of the ALDO. As a result, the overall PSR is maintained below -10dB at all frequencies. The switch M_{S1} is used in parallel with R_1 to pull the gate voltage of the native NMOSs, V_{RC} , up to V_{IN} briefly at each beginning of $S2$ or $S3$ (S_{RC}) preventing a sudden voltage drop at V_{RC} when I_L occurs momentarily. The switch M_{S2} is used to pull V_{OUT} down to ground briefly after the end of each $S3$ (S_{OVST}), effectively mitigating a sudden overshoot at V_{OUT} when i_L decreases momentarily.

The bottom of Fig. 8.5.3 shows the implementation of the I_D -adaptor, which includes two adaptation circuits for $I_{D,ACT}$ and $I_{D,TOG}$. Each circuit has a dead-zone (DZ)-comparator and an accumulator (ACC). To detect ΔV_{DR00P} each DZ-comparator samples V_{OUT} at every rising edge (RE) of the sampling clocks, i.e., $s_{SP,ACC}$ or $s_{SP,TOG}$, which are generated by delaying s_{ACT} or S_{TOG} . According to the outputs of the DZ-comparators ($D_{DZ,A}$ and $D_{DZ,T}$), the codes of the ACCs (D_{ACT} and D_{TOG}) are updated until the sampled V_{OUT} is within $V_{\text{REF}} \pm 10\text{mV}$ (V_{REF} is the reference voltage). The top of Fig. 8.5.3 shows that V_{OUT} initially exhibits significant ΔV_{DR00P} since I_L cannot be compensated only with an ALDO. However, after adaptation, I_L is compensated accurately by optimal $I_{D,ACT}$ and $I_{D,TOG}$, resulting in minimal ΔV_{DR00P} and a short T_S . For real products, D_{ACT} and D_{TOG} are initially adapted during the training time and then updated in the background. The initial adaptation takes less than 200ns.

The proposed CA-HLDO occupies 0.014mm^2 area in 40nm CMOS (Fig. 8.5.7). The top of Fig. 8.5.4 depicts the measurement scenario: $S2$ is maintained for 4.8ns , followed by three sequential $S3$ events, where WDQS/ 2_{IN} toggles for 20ns each with a 2ns gap of $S4$. The bottom of Fig. 8.5.4 shows the load transient response and the resultant jitter histogram of the output clock (S_{MEAS}) at 2.5GHz . When D_{ACT} and D_{TOG} were set to zero, the limited BW of the ALDO caused a massive ΔV_{DR00P} of 185.0mV and a substantial jitter p-p of 52.0ps (left). (No AM modulation was observed in s_{MEAS} due to the measurement buffer using a constant V_{DD} .) However, with optimized adaptation of D_{ACT} and D_{TOG} , I_D quickly and accurately compensated for I_L , reducing ΔV_{DROOP} and jitter_{p-p} to 8.7mV and 10.8ps , respectively (right). The overshoot in every $S4$ was reduced sufficiently before the next $S3$ began. The top of Fig. 8.5.5 shows that the proposed CA-HLDO remains effective even with variations in V_{IN} and V_{OUT} maintaining ΔV_{DROOF} below 10mV in 40ops . This validates the I_L adaptation, which continuously calibrates I_D in response to changes in

I_L . When V_{OUT} was increased to 0.85V, resulting in a larger I_L , the maximum ΔV_{DROOP} slightly exceeded 10mV but was reduced to below 10mV within 400ps. The bottom of Fig. 8.5.5 shows that ΔV_{DROOP} and $PSIJ$ remained minimal across a wide range of S_{MEAS} frequencies. Figure 8.5.6 shows the performance comparison of the proposed CA-HLDO with the prior-art LDOs. Employing the proposed command-aware scheme with the I_D -adaptation, it achieved significantly smaller ΔV_{DROOP} and T_s even with a shorter T_{EDGE} . Consequently, it achieved the best FoMs while using $150\mu A / Q (V_{IN} = 1.05V)$ and $20pF C_{OUT}$. It also provides a PSR of at least -10dB across all frequencies.



Figure 8.5.1:
Psij issues of wdqs buffer in hbm, ldo requirements, and proposed command-aware hybrid ldo (ca-hldo) (left); il profile and different magnitudes of voltage droop (Δv_{droop}) and psij according to types of ldos (right).



Figure 8.5.2:
Overall architecture of the proposed ca-hldo(top); configuration of the wdqs buffer (bottom left); and overall psr of the proposed ca-hldo (bottom right).



Figure 8.5.3:
Il profile and its compensation by id & ia before and after ld-adaptation (top); implementation of ld-adaptor (bottom).



Figure 8.5.4:
Measured load transient response and jitter histogram of smeas at 2.5ghz when dact and dtogwere set to zero (left); and when dact and dtogwere optimized (right).



Figure 8.5.5:
Measured load transient response and jitter histogram of smeasat 2.5ghz when voutis 0.65 and 0.85v (top); measured load transient response and jitten histogram of smeaswith frequencies of 1.5 and 1.0ghz (bottom).



Figure 8.5.6:
Performance comparison with prior-art ldos for small Δv_{droop} .



Figure 8.5.7:
Die micrograph

ACKNOWLEDGEMENT

This work was supported in part by Samsung Electronics Co. Ltd., in part by the National R&D Program through the NRF funded by the Ministry of Science and ICT under Grant 2020R1A2C2004260, and in part by the || PT through the Artificial Intelligence Semiconductor Support Program to Nurture the Best Talents Grant funded by the MSIT under Grant IITP-2023-RS-2023-00256081.

IEEE Personal Account

CHANGE USERNAME/PASSWORD

Purchase Details

PAYMENT OPTIONS

VIEW PURCHASED DOCUMENTS

Profile Information

COMMUNICATIONS PREFERENCES

PROFESSION AND EDUCATION

TECHNICAL INTERESTS

Need Help?

US & CANADA: +1 800 678 4333

WORLDWIDE: +1 732 981 0060

CONTACT & SUPPORT

Follow

[f](#) [@](#) [in](#) [v](#) [X](#)

About IEEE *Xplore* | Contact Us | Help | Accessibility | Terms of Use | Nondiscrimination Policy | IEEE Ethics Reporting [🔗](#) | Sitemap | IEEE Privacy Policy

A public charity, IEEE is the world's largest technical professional organization dedicated to advancing technology for the benefit of humanity.

© Copyright 2026 IEEE - All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies.